between BSTs and heaps, abiding by BST's invariants for keys, and heap's invariants for priorities.

- **Randomized Treap**: if we randomly assign priorities, drawing them from a uniform continuous set (like, but not limited to, all real numbers between 0 and 1), we can mathematically guarantee that, for large enough values of `n`, the tree will store `n` elements and maintain a height not greater than `2*log(n)`.
- **Profiling:** besides the theoretical guarantees, it's possible to verify (as we did with Java implementations) that randomized treaps will keep a logarithmic height even in the worst-case scenarios for BSTs.
- **Performance:** moreover, the performance in term of CPU running time and memory usage is comparable for both data structures in the general case, and much better for randomized treaps in the edge cases where BSTs struggle.

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[1] Treaps were introduced in "Aragon, Cecilia R., and Raimund G. Seidel. "Randomized search trees." 30th Annual Symposium on Foundations of Computer Science. IEEE, 1989". Although the title of the paper might be misleading, we will see later in this chapter how treaps are related to randomized search trees.

[2] A portmanteau is a blend of two or more words, where parts of each word are combined into a new word.

[3] Red-black and 2-3 trees are fancy versions of balanced BSTs. We'll talk more about them in a few sections.

[4] You can read more about the issue with stack overflow and tail call optimization in appendix E.

[5] In our example, the lowest priority corresponds to the highest availability in stock, and so +∞ is the highest possible value for the units in stock.

[6] Aho, Alfred V., and John E. Hopcroft. The design and analysis of computer algorithms. Pearson Education India, 1974.

[7] Guibas, Leo J., and Robert Sedgewick. "A dichromatic framework for balanced trees." 19th Annual Symposium on Foundations of Computer Science (sfcs 1978). IEEE, 1978.

[8] Meant as: balanced, with a high probability.

[9] See appendix C for a recap on hashing.

[10] It goes without saying that this holds if the pseudo random generators are implemented properly, and anyway within the limits caused by the fact that classic computer can't offer true randomness.